80 FM/FIM collinear spin layer groups

Magnetic-axis directions, SOC magnetic layer groups and layer-resolved intrinsic nonlinear Hall constraints

Coverage. All 80 entries of Table S3, in the original order, expanded into **540 direction branches**: 2 groups have 1 branch, 2 have 2, 23 have 3, 30 have 7, 6 have 8, 12 have 11, and 5 have 15. Directions related by a parent symmetry are retained in the fixed source coordinate setting so that the displayed tensor indices stay meaningful. These 540 rows are direction strata, not 540 distinct MLG types.

Source. J.-X. Du, S. Zeng and Y.-J. Zhao, *Stacking theory for bilayer two-dimensional magnets*, Phys. Rev. B **113**, 214455 (2026), DOI: 10.1103/g471-xrkl; Supplemental Material, Table S3, pp. 21–33. The source supplies the cSLGs and operations. The magnetic-axis branches, SOC MLGs and response constraints in this report are calculated results.

Magnetic axis. Here $\hat{\mathbf{m}}$ is the common collinear magnetic axis, which can be chosen along the net magnetization when it is nonzero. For ferrimagnets, sublattice moments keep their relative signs and magnitudes while the common axis is rotated. This definition also covers compensated ferrimagnets; a nonzero net moment is not assumed. The cSLG alone does not distinguish FM from FIM or determine the moment magnitudes. The direction column therefore refers to the magnetic axis rather than a Néel vector.

Direction notation. $\mathrm{AX}_{[uvw]}$ denotes a special axis; $\mathrm{PL}_{(hkl)}$ denotes a direction in a special plane excluding all axes lying in that plane; GEN denotes a direction outside every listed axis and plane. Only symmetry-distinct stabilizer strata of a given group are introduced; for example, groups 1.1.1 and 2.1.4 have a single GEN row that covers all directions; groups 65.1.462 and 66.1.464 need only $\mathrm{AX}_{[001]}$ and GEN.

Coordinates. Tensor components are Cartesian, with z normal to the layer. For rectangular and tetragonal groups, $x \parallel \mathbf{a}$, $y \parallel \mathbf{b}$. For hexagonal groups, $\mathbf{a} = (1, 0, 0)$ and $\mathbf{b} = (-1/2, \sqrt{3}/2, 0)$, while Cartesian $y \parallel \hat{z} \times \mathbf{a}$. Miller indices refer to these crystallographic bases. Thus [010] is not the Cartesian y direction in the hexagonal setting. Hexagonal in-plane operations are displayed with explicit crystallographic directions; $M_{[uvw]}$ has its Cartesian normal along $[uvw]$. The source triplets $\{x, y, xy\}$ and $\{1, 2, 3\}$ correspond as sets to $\{[100], [010], [110]\}$ and $\{[1\bar{1}0], [120], [210]\}$. No individual identification of the source's numbered subscripts is needed.

Operation columns. Translation parts are omitted and repeated point operations are merged. All distinct nontrivial-part SLG point operations are shown on separate lines; the common spin-only factor is implicit. Every MLG row explicitly lists its point operations on one line. Full MLG symbols and numbers are nevertheless identified from the full affine operations in the source, then verified by a basis/origin change preserving the layer normal. MLG symbols use their standard settings; the explicitly listed operations and tensor indices remain in the source coordinate setting. An affine witness for every row is included in the data package. Omitting translations from the displayed columns does not redefine these group identifiers.

Tensor convention. Set $A^\ell = \chi^\ell_{xyy}$ and $B^\ell = \chi^\ell_{yxx}$ for $\ell = t, b$, with $\chi_{abc} = -\chi_{bac}$ and in-plane indices. Then $\chi_{xyx} = -B$, $\chi_{yxy} = -A$, and all other in-plane entries vanish. The two-component vector $\mathbf{v}^\ell = (A^\ell, B^\ell)$ transforms as $\mathbf{v}^{g\ell} = q_g R_\parallel \mathbf{v}^\ell$. This statement is specific to the $\mathcal{T}$-odd intrinsic Hall part.

Mark	Constraint on the displayed component	Meaning
$+$	$\chi^t = \chi^b$, allowed	Equal layer contributions
$-$	$\chi^t = -\chi^b$, allowed	Hidden, layer compensated
0	$\chi^t = \chi^b = 0$	Forbidden in each layer
v	No fixed proportional relation between the same component in the two layers	See coupled constraints when starred

A star on a direction indicates additional relations between A and B, listed completely in Appendix A (112 branches). In these rows, v must not be read as absence of all symmetry relations: cross-component relations may remain. All allowed values can accidentally vanish microscopically.

Assumptions and calculation. Fixed structure and collinear order; two physical layers at opposite sides of $z = 0$. All nontrivial-part spin labels are $s_g = +1$. The SOC subgroup satisfies $q_g \det(R_g) R_g \hat{\mathbf{m}} = \hat{\mathbf{m}}$, with $q_g = +1$ for g and $q_g = -1$ for $g\mathcal{T}$. Operations with $R_{zz} = -1$ exchange the two layers. Branches are constructed from all real ± 1 eigenspaces and their intersections. These are symmetry-allowed constrained orientations, not predictions of magnetic easy axes or stability.

Source audit (S1). Target 77.1.505 (Table S3, p. 32), operation 09, is printed as $\{-1\|M_y\}$. This contradicts the FM/FIM classification, the all-+1 spin labels in its symbol, and group closure: two +1 operations cannot produce an isolated −1 operation. We use $\{+1\|M_y\}$; all 80 target operation lists then close and agree with the printed group symbols. The original operation is retained in the raw data. This is a consistency repair, not an author-issued erratum.

The MLG reference catalog additionally retains two previously documented consistency repairs: 31.5.192, operation 04, $s_{M_x} = +1$; and 39.5.260, operation 08, translation $(0, 1/2, 0)$. These two records are outside the 80 FM/FIM targets.

Validation. All 540 MLG assignments are checked against complete affine operations. Each allowed response space is independently checked using the full third-rank Hall tensor, and direction coverage is sampled on every axis and plane as well as generic orientations.

Complete table, in Table S3 order

First three columns are merged across all direction branches of one cSLG. Asterisks refer to Appendix A; S1 refers to the source audit on the first page.

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
01 1.1.1 $p^1 1$ SI p. 21	FM/FIM	$\{1\|E\}$	GEN	$p1$ 1.1.1	$\{E\}$	v	v
02 2.1.4 $p^1\bar{1}$ SI p. 21	FM/FIM	$\{1\|E\}$ $\{1\|\mathcal{P}\}$	GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
03 3.1.8 $p^1 1^1 1^1 2$ SI p. 21	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$	$\mathrm{AX}_{[001]}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
04 4.1.12 $p^1 1^1 1^1 m$ SI p. 21	FM/FIM	$\{1\|E\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p11m$ 4.1.12	$\{E, M_z\}$	+	+
			$\mathrm{PL}_{(001)}$	$p11m'$ 4.3.14	$\{E, M_z\mathcal{T}\}$	–	–
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
05 5.1.17 $p^1 1^1 1^1 a$ SI p. 21	FM/FIM	$\{1\|E\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p11a$ 5.1.17	$\{E, M_z\}$	+	+
			$\mathrm{PL}_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
06 6.1.21 $p^1 1^1 1^1 2/^1 m$ SI p. 21	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p112/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
07 7.1.28 $p^1 1^1 1^1 2/^1 a$ SI p. 21	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p112/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
08 8.1.34 $p^1 2^1 1^1 1$ SI p. 21	FM/FIM	$\{1\|E\}$ $\{1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p211$ 8.1.34	$\{E, C_{2x}\}$	+	–
			$\mathrm{PL}_{(100)}$	$p2'11$ 8.3.36	$\{E, C_{2x}\mathcal{T}\}$	–	+
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
09 9.1.41 $p^1 2_1^1 1^1 1$ SI p. 21	FM/FIM	$\{1\|E\}$ $\{1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p2_111$ 9.1.41	$\{E, C_{2x}\}$	+	–
			$\mathrm{PL}_{(100)}$	$p2'_111$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	–	+
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
10 10.1.45 $c^12^11^11$ SI p. 21	FM/FIM	$\{1\|\|E\}$ $\{1\|\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$c211$ 10.1.45	$\{E,\, C_{2x}\}$	$+$	$-$
			$\mathrm{PL}_{(100)}$	$c2'11$ 10.3.47	$\{E,\, C_{2x}\mathcal{T}\}$	$-$	$+$
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
11 11.1.50 $p^1m^11^11$ SI p. 21	FM/FIM	$\{1\|\|E\}$ $\{1\|\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm11$ 11.1.50	$\{E,\, M_x\}$	0	v
			$\mathrm{PL}_{(100)}$	$pm'11$ 11.3.52	$\{E,\, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
12 12.1.57 $p^1b^11^11$ SI p. 21	FM/FIM	$\{1\|\|E\}$ $\{1\|\|M_x\}$	$\mathrm{AX}_{[100]}$	$pb11$ 12.1.57	$\{E,\, M_x\}$	0	v
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E,\, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
13 13.1.61 $c^1m^11^11$ SI p. 21	FM/FIM	$\{1\|\|E\}$ $\{1\|\|M_x\}$	$\mathrm{AX}_{[100]}$	$cm11$ 13.1.61	$\{E,\, M_x\}$	0	v
			$\mathrm{PL}_{(100)}$	$cm'11$ 13.3.63	$\{E,\, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
14 14.1.66 $p^12/^1m^11^11$ SI p. 21	FM/FIM	$\{1\|\|E\}$ $\{1\|\|C_{2x}\}$ $\{1\|\|\mathcal{P}\}$ $\{1\|\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2/m11$ 14.1.66	$\{E,\, C_{2x},\, \mathcal{P},\, M_x\}$	0	$-$
			$\mathrm{PL}_{(100)}$	$p2'/m'11$ 14.5.70	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T}\}$	$-$	0
			GEN	$p\bar{1}$ 2.1.4	$\{E,\, \mathcal{P}\}$	$-$	$-$
15 15.1.76 $p^12_1/^1m^11^11$ SI p. 21	FM/FIM	$\{1\|\|E\}$ $\{1\|\|C_{2x}\}$ $\{1\|\|\mathcal{P}\}$ $\{1\|\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2_1/m11$ 15.1.76	$\{E,\, C_{2x},\, \mathcal{P},\, M_x\}$	0	$-$
			$\mathrm{PL}_{(100)}$	$p2'_1/m'11$ 15.4.79	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T}\}$	$-$	0
			GEN	$p\bar{1}$ 2.1.4	$\{E,\, \mathcal{P}\}$	$-$	$-$
16 16.1.83 $p^12/^1b^11^11$ SI p. 22	FM/FIM	$\{1\|\|E\}$ $\{1\|\|C_{2x}\}$ $\{1\|\|\mathcal{P}\}$ $\{1\|\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2/b11$ 16.1.83	$\{E,\, C_{2x},\, \mathcal{P},\, M_x\}$	0	$-$
			$\mathrm{PL}_{(100)}$	$p2'/b'11$ 16.4.86	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T}\}$	$-$	0
			GEN	$p\bar{1}$ 2.1.4	$\{E,\, \mathcal{P}\}$	$-$	$-$
17 17.1.90 $p^12_1/^1b^11^11$ SI p. 22	FM/FIM	$\{1\|\|E\}$ $\{1\|\|C_{2x}\}$ $\{1\|\|\mathcal{P}\}$ $\{1\|\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2_1/b11$ 17.1.90	$\{E,\, C_{2x},\, \mathcal{P},\, M_x\}$	0	$-$
			$\mathrm{PL}_{(100)}$	$p2'_1/b'11$ 17.4.93	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T}\}$	$-$	0
			GEN	$p\bar{1}$ 2.1.4	$\{E,\, \mathcal{P}\}$	$-$	$-$

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
18 18.1.95 $c^12/^1m^11^11$ SI p. 22	FM/FIM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$	$AX_{[100]}$	$c2/m11$ 18.1.95	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	−
			$PL_{(100)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	−	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	−	−
19 19.1.104 $p^12^12^12$ SI p. 22	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$	$AX_{[100]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	+	−
			$AX_{[010]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	−	+
			$AX_{[001]}$	$p2'2'2$ 19.3.106	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$PL_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	+	−
			$PL_{(100)}$	$p2'11$ 8.3.36	$\{E, C_{2x}\mathcal{T}\}$	−	+
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
20 20.1.111 $p^12_1^12^12$ SI p. 22	FM/FIM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2z}\}$	$AX_{[100]}$	$p2_12'2'$ 20.3.113	$\{E, C_{2x}, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}\}$	+	−
			$AX_{[010]}$	$p2'_122'$ 20.4.114	$\{E, C_{2x}\mathcal{T}, C_{2y}, C_{2z}\mathcal{T}\}$	−	+
			$AX_{[001]}$	$p2'_12'2$ 20.5.115	$\{E, C_{2x}\mathcal{T}, C_{2y}\mathcal{T}, C_{2z}\}$	0	0
			$PL_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$PL_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	+	−
			$PL_{(100)}$	$p2'_111$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	−	+
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
21 21.1.118 $p^12_1^12_1^12$ SI p. 22	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$	$AX_{[100]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	+	−
			$AX_{[010]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	−	+
			$AX_{[001]}$	$p2'_12'_12$ 21.3.120	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$PL_{(010)}$	$p2'_111$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	+	−
			$PL_{(100)}$	$p2'_111$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	−	+
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
22 22.1.122 $c^12^12^12$ SI p. 22	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$	$AX_{[100]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	+	−
			$AX_{[010]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	−	+
			$AX_{[001]}$	$c2'2'2$ 22.3.124	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$PL_{(010)}$	$c2'11$ 10.3.47	$\{E, C_{2y}\mathcal{T}\}$	+	−
			$PL_{(100)}$	$c2'11$ 10.3.47	$\{E, C_{2x}\mathcal{T}\}$	−	+
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
23 23.1.129 $p^1m^1m^12$ SI p. 22	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$AX_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$AX_{[001]}$	$pm'm'2$ 23.3.131	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$PL_{(010)}$	$pm'11$ 11.3.52	$\{E, M_y\mathcal{T}\}$	0	v
			$PL_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
24 24.1.136 $p^1m^1a^12$ SI p. 22	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pma'2'$ 24.5.140	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$AX_{[010]}$	$pm'a2'$ 24.4.139	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$AX_{[001]}$	$pm'a'2$ 24.3.138	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$PL_{(010)}$	$pb'11$ 12.3.59	$\{E, M_y\mathcal{T}\}$	0	v
			$PL_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
25 25.1.143 $p^1b^1a^12$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$AX_{[010]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$AX_{[001]}$	$pb'a'2$ 25.3.145	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$PL_{(010)}$	$pb'11$ 12.3.59	$\{E, M_y\mathcal{T}\}$	0	v
			$PL_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
26 26.1.147 $c^1m^1m^12$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$AX_{[010]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$AX_{[001]}$	$cm'm'2$ 26.3.149	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$PL_{(010)}$	$cm'11$ 13.3.63	$\{E, M_y\mathcal{T}\}$	0	v
			$PL_{(100)}$	$cm'11$ 13.3.63	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
27 27.1.154 $p^1m^12^1m$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$AX_{[100]}$	$pm2'm'$ 27.4.157	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$AX_{[010]}$	$pm'2m'$ 27.3.156	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$AX_{[001]}$	$pm'2'm$ 27.5.158	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$PL_{(001)}$	$p11m'$ 4.3.14	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$PL_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
28 28.1.167 $p^1m^12_1^1b$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pm2_1'b'$ 28.5.171	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$AX_{[010]}$	$pm'2_1b'$ 28.3.169	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$AX_{[001]}$	$pm'2_1'b$ 28.4.170	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}\}$	+	0
			$PL_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$p2_1'11$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$PL_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
29 29.1.174 $p^1b^12_1^1m$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$AX_{[100]}$	$pb2_1'm'$ 29.4.177	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$AX_{[010]}$	$pb'2_1m'$ 29.3.176	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$AX_{[001]}$	$pb'2_1'm$ 29.5.178	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$PL_{(001)}$	$p11m'$ 4.3.14	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$p2_1'11$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$PL_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
30 30.1.181 $p^1b^12^1b$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pb2'b'$ 30.5.185	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pb'2b'$ 30.3.183	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pb'2'b$ 30.4.184	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
31 31.1.188 $p^1m^12^1a$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm2'a'$ 31.5.192	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pm'2a'$ 31.3.190	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pm'2'a$ 31.4.191	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$\mathrm{PL}_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
32 32.1.197 $p^1m^12_1^1n$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm2_1'n'$ 32.5.201	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pm'2_1n'$ 32.3.199	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pm'2_1'n$ 32.4.200	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1'11$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$\mathrm{PL}_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
33 33.1.202 $p^1b^12_1^1a$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pb2_1'a'$ 33.5.206	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pb'2_1a'$ 33.3.204	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pb'2_1'a$ 33.4.205	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1'11$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
34 34.1.207 $p^1b^12^1n$ SI p. 23	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pb2'n'$ 34.4.210	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[010]}$	$pb'2n'$ 34.3.209	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pb'2'n$ 34.5.211	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$\mathrm{PL}_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
35 35.1.212 $c^1m^12^1m$ SI p. 24	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$cm2'm'$ 35.4.215	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[010]}$	$cm'2m'$ 35.3.214	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$cm'2'm$ 35.5.216	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$\mathrm{PL}_{(001)}$	$p11m'$ 4.3.14	$\{E, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$c2'11$ 10.3.47	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$\mathrm{PL}_{(100)}$	$cm'11$ 13.3.63	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
36 36.1.221 $c^1m^12^1a$ SI p. 24	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$cm2'a'$ 36.4.224	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[010]}$	$cm'2a'$ 36.3.223	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$cm'2'a$ 36.5.225	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$\mathrm{PL}_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$c2'11$ 10.3.47	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$\mathrm{PL}_{(100)}$	$cm'11$ 13.3.63	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
37 37.1.230 $p^1m^1m^1m$ SI p. 24	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pm'm'm$ 37.4.233	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
38 38.1.243 $p^1m^1a^1a$ SI p. 24	FM/FIM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pma'a'$ 38.4.246	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_x, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[010]}$	$pm'aa'$ 38.6.248	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\}$	–	0
			$\mathrm{AX}_{[001]}$	$pm'a'a$ 38.5.247	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, M_z, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
39 39.1.256 $p^1b^1a^1n$ SI p. 24	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pb'a'n$ 39.4.259	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
40 40.1.263 $p^1m^1a^1m$ SI p. 24	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pma'm'$ 40.6.268	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pm'am'$ 40.4.266	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_y, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pm'a'm$ 40.5.267	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
41 41.1.276 $p^1m^1m^1a$ SI p. 25	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmm'a'$ 41.6.281	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pm'ma'$ 41.5.280	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pm'm'a$ 41.4.279	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
42 42.1.289 $p^1m^1a^1n$ SI p. 25	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pma'n'$ 42.6.294	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pm'an'$ 42.4.292	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_y, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pm'a'n$ 42.5.293	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
43 43.1.298 $p^1b^1a^1a$ SI p. 25	FM/FIM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pba'a'$ 43.4.301	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_x, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[010]}$	$pb'aa'$ 43.5.302	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\}$	–	0
			$\mathrm{AX}_{[001]}$	$pb'a'a$ 43.6.303	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, M_z, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
44 44.1.307 $p^1b^1a^1m$ SI p. 25	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pb'a'm$ 44.4.310	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pb'a'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
45 45.1.314 $p^1b^1m^1a$ SI p. 25	FM/FIM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pbm'a'$ 45.5.318	$\{E, C_{2y}\mathcal{T}, C_{2x}, C_{2z}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[010]}$	$pb'ma'$ 45.4.317	$\{E, C_{2y}, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}, M_y, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pb'm'a$ 45.6.319	$\{E, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2z}, \mathcal{P}, M_y\mathcal{T}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
46 46.1.323 $p^1m^1m^1n$ SI p. 26	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$pm'm'n$ 46.4.326	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
47 47.1.330 $c^1m^1m^1m$ SI p. 26	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$cm'm'm$ 47.4.333	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
48 48.1.343 $c^1m^1m^1e$ SI p. 26	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[001]}$	$cm'm'e$ 48.4.346	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
49 49.1.354 p^14 SI p. 26	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$	$\mathrm{AX}_{[001]}$	$p4$ 49.1.354	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
50 50.1.358 $p^1\bar{4}$ SI p. 27	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$	$\mathrm{AX}_{[001]}$	$p\bar{4}$ 50.1.358	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}^{-1}_{4z}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
51 51.1.362 $p^14/^1m$ SI p. 27	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p4/m$ 51.1.362	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}_{4z}^{-1}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
52 52.1.369 $p^14/^1n$ SI p. 27	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p4/n$ 52.1.369	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}_{4z}^{-1}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
53 53.1.374 $p^14^12^12$ SI p. 27	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	+	–
			$\mathrm{AX}_{[010]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	–	+
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p42'2'$ 53.3.376	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	+	–
			$\mathrm{PL}_{(100)}$	$p2'11$ 8.3.36	$\{E, C_{2x}\mathcal{T}\}$	–	+
			$\mathrm{PL}^*_{(110)}$	$c2'11$ 10.3.47	$\{E, C_{2xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
54 54.1.381 $p^14^12_12$ SI p. 27	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	AX$_{[100]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	+	−
			AX$_{[010]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	−	+
			AX$^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			AX$^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			AX$_{[001]}$	$p42'_12'$ 54.3.383	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			PL$_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			PL$_{(010)}$	$p2'_111$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	+	−
			PL$_{(100)}$	$p2'_111$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	−	+
			PL$^*_{(110)}$	$c2'11$ 10.3.47	$\{E, C_{2xy}\mathcal{T}\}$	v	v
			PL$^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
55 55.1.386 $p^14^1m^1m$ SI p. 27	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	AX$_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			AX$_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			AX$^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			AX$^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			AX$_{[001]}$	$p4m'm'$ 55.3.388	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			PL$_{(001)}$	$p11\bar{2}'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			PL$_{(010)}$	$pm'11$ 11.3.52	$\{E, M_y\mathcal{T}\}$	0	v
			PL$_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			PL$^*_{(110)}$	$cm'11$ 13.3.63	$\{E, M_{xy}\mathcal{T}\}$	v	v
			PL$^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E, M_{\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
56 56.1.393 $p^14^1b^1m$ SI p. 28	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'2'$ 25.4.146	$\{E,\ C_{2z}\mathcal{T},\ M_y\mathcal{T},\ M_x\}$	0	v
			$\mathrm{AX}_{[010]}$	$pba'2'$ 25.4.146	$\{E,\ C_{2z}\mathcal{T},\ M_y,\ M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E,\ C_{2z}\mathcal{T},\ M_{xy},\ M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E,\ C_{2z}\mathcal{T},\ M_{xy}\mathcal{T},\ M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4b'm'$ 56.3.395	$\{E,\ C_{2z},\ C_{4z},\ C^{-1}_{4z},\ M_y\mathcal{T},\ M_x\mathcal{T},\ M_{xy}\mathcal{T},\ M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E,\ C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb'11$ 12.3.59	$\{E,\ M_y\mathcal{T}\}$	0	v
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E,\ M_x\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E,\ M_{xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
57 57.1.398 $p^1\bar{4}^12^1m$ SI p. 28	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p22'2'$ 19.4.107	$\{E,\ C_{2z}\mathcal{T},\ C_{2y}\mathcal{T},\ C_{2x}\}$	$+$	$-$
			$\mathrm{AX}_{[010]}$	$p22'2'$ 19.4.107	$\{E,\ C_{2z}\mathcal{T},\ C_{2y},\ C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E,\ C_{2z}\mathcal{T},\ M_{xy},\ M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E,\ C_{2z}\mathcal{T},\ M_{xy}\mathcal{T},\ M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}2'm'$ 57.3.400	$\{E,\ C_{2z},\ \bar{C}_{4z},\ \bar{C}^{-1}_{4z},\ C_{2y}\mathcal{T},\ C_{2x}\mathcal{T},\ M_{xy}\mathcal{T},\ M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E,\ C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E,\ C_{2y}\mathcal{T}\}$	$+$	$-$
			$\mathrm{PL}_{(100)}$	$p2'11$ 8.3.36	$\{E,\ C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E,\ M_{xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
58 58.1.405 $p^1\bar{4}^1 2_1^1 m$ SI p. 28	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p2_12_1'2'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}_{[010]}$	$p2_12_1'2'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}2_1'm'$ 58.3.407	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2_1'11$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	$+$	$-$
			$\mathrm{PL}_{(100)}$	$p2_1'11$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E, M_{xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E, M_{\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
59 59.1.410 $p^1\bar{4}^1 m^1 2$ SI p. 28	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}m'2'$ 59.3.412	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y\mathcal{T}, M_x\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pm'11$ 11.3.52	$\{E, M_y\mathcal{T}\}$	0	v
			$\mathrm{PL}_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(110)}$	$c2'11$ 10.3.47	$\{E, C_{2xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
60 60.1.417 $p^1\bar{4}^1b^12$ SI p. 28	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'2'$ 25.4.146	$\{E,\, C_{2z}\mathcal{T},\, M_y\mathcal{T},\, M_x\}$	0	v
			$\mathrm{AX}_{[010]}$	$pba'2'$ 25.4.146	$\{E,\, C_{2z}\mathcal{T},\, M_y,\, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E,\, C_{2z}\mathcal{T},\, C_{2xy},\, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E,\, C_{2z}\mathcal{T},\, C_{2xy}\mathcal{T},\, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}b'2'$ 60.3.419	$\{E,\, C_{2z},\, \bar{C}_{4z},\, \bar{C}^{-1}_{4z},\, M_y\mathcal{T},\, M_x\mathcal{T},\, C_{2xy}\mathcal{T},\, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E,\, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb'11$ 12.3.59	$\{E,\, M_y\mathcal{T}\}$	0	v
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E,\, M_x\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(110)}$	$c2'11$ 10.3.47	$\{E,\, C_{2xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E,\, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
61 61.1.422 $p^14/^1m^1m^1m$ SI p. 28	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'm'$ 37.5.234	$\{E,\, C_{2z}\mathcal{T},\, C_{2y}\mathcal{T},\, C_{2x},\, \mathcal{P},\, M_z\mathcal{T},\, M_y\mathcal{T},\, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pmm'm'$ 37.5.234	$\{E,\, C_{2z}\mathcal{T},\, C_{2y},\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_z\mathcal{T},\, M_y,\, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}^*_{[110]}$	$cmm'm'$ 47.5.334	$\{E,\, C_{2z}\mathcal{T},\, C_{2xy}\mathcal{T},\, C_{2\bar{x}y},\, \mathcal{P},\, M_z\mathcal{T},\, M_{xy}\mathcal{T},\, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E,\, C_{2z}\mathcal{T},\, C_{2xy}\mathcal{T},\, C_{2\bar{x}y},\, \mathcal{P},\, M_z\mathcal{T},\, M_{xy}\mathcal{T},\, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4/mm'm'$ 61.4.425	$\{E,\, C_{2z},\, C_{4z},\, C^{-1}_{4z},\, C_{2y}\mathcal{T},\, C_{2x}\mathcal{T},\, C_{2xy}\mathcal{T},\, C_{2\bar{x}y}\mathcal{T},\, \mathcal{P},\, M_z,\, \bar{C}_{4z},\, \bar{C}^{-1}_{4z},\, M_y\mathcal{T},\, M_x\mathcal{T},\, M_{xy}\mathcal{T},\, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E,\, C_{2z}\mathcal{T},\, \mathcal{P},\, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'/m'11$ 14.5.70	$\{E,\, C_{2y}\mathcal{T},\, \mathcal{P},\, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'/m'11$ 14.5.70	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{P},\, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E,\, C_{2xy}\mathcal{T},\, \mathcal{P},\, M_{xy}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E,\, C_{2\bar{x}y}\mathcal{T},\, \mathcal{P},\, M_{\bar{x}y}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E,\, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
62 62.1.435 $p^14/^1n^1b^1m$ SI p. 29	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}^*_{[110]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4/nb'm'$ 62.4.438	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}, M_{xy}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
63 63.1.444 $p^14/^1m^1b^1m$ SI p. 29	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}^*_{[110]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4/mb'm'$ 63.4.447	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1'/b'11$ 17.4.93	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2_1'/b'11$ 17.4.93	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}, M_{xy}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
64 64.1.453 $p^14/^1n^1m^1m$ SI p. 30	FM/FIM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}^*_{[110]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4/nm'm'$ 64.4.456	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}, M_{xy}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
65 65.1.462 p^13 SI p. 30	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C^{-1}_{3z}\}$	$\mathrm{AX}_{[001]}$	$p3$ 65.1.462	$\{E, C_{3z}, C^{-1}_{3z}\}$	0	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
66 66.1.464 $p^1\bar{3}$ SI p. 30	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C^{-1}_{3z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}^{-1}_{3z}\}$	$\mathrm{AX}_{[001]}$	$p3$ 66.1.464	$\{E, C_{3z}, C^{-1}_{3z}, \mathcal{P}, \bar{C}_{3z}, \bar{C}^{-1}_{3z}\}$	0	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
67 67.1.467 $p^13^11^12$ SI p. 30	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C^{-1}_{3z}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}^*_{[1\bar{1}0]}$	$c211$ 10.1.45	$\{E, C_{2,[1\bar{1}0]}\}$	v	v
			$\mathrm{AX}_{[120]}$	$c211$ 10.1.45	$\{E, C_{2,[120]}\}$	–	+
			$\mathrm{AX}^*_{[210]}$	$c211$ 10.1.45	$\{E, C_{2,[210]}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p312'$ 67.3.469	$\{E, C_{3z}, C^{-1}_{3z}, C_{2,[210]}\mathcal{T}, C_{2,[120]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2'11$ 10.3.47	$\{E, C_{2,[120]}\mathcal{T}\}$	+	–
			$\mathrm{PL}^*_{(100)}$	$c2'11$ 10.3.47	$\{E, C_{2,[210]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
68 68.1.470 $p^13^12^11$ SI p. 30	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$	$\mathrm{AX}_{[100]}$	$c211$ 10.1.45	$\{E,\ C_{2,[100]}\}$	+	−
			$\mathrm{AX}^*_{[010]}$	$c211$ 10.1.45	$\{E,\ C_{2,[010]}\}$	v	v
			$\mathrm{AX}^*_{[110]}$	$c211$ 10.1.45	$\{E,\ C_{2,[110]}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p32'1$ 68.3.472	$\{E,\ C_{3z},\ C_{3z}^{-1},\ C_{2,[110]}\mathcal{T},\ C_{2,[100]}\mathcal{T},\ C_{2,[010]}\mathcal{T}\}$	0	0
			$\mathrm{PL}^*_{(110)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[110]}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[100]}\mathcal{T}\}$	−	+
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[010]}\mathcal{T}\}$	v	v
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
69 69.1.473 $p^13^1m^11$ SI p. 30	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$	$\mathrm{AX}_{[100]}$	$cm11$ 13.1.61	$\{E,\ M_{[100]}\}$	0	v
			$\mathrm{AX}^*_{[010]}$	$cm11$ 13.1.61	$\{E,\ M_{[010]}\}$	v	v
			$\mathrm{AX}^*_{[110]}$	$cm11$ 13.1.61	$\{E,\ M_{[110]}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p3m'1$ 69.3.475	$\{E,\ C_{3z},\ C_{3z}^{-1},\ M_{[110]}\mathcal{T},\ M_{[100]}\mathcal{T},\ M_{[010]}\mathcal{T}\}$	0	0
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E,\ M_{[110]}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{[100]}\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{[010]}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
70 70.1.476 $p^13^11^1m$ SI p. 31	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}^*_{[1\bar{1}0]}$	$cm11$ 13.1.61	$\{E,\ M_{[1\bar{1}0]}\}$	v	v
			$\mathrm{AX}_{[120]}$	$cm11$ 13.1.61	$\{E,\ M_{[120]}\}$	v	0
			$\mathrm{AX}^*_{[210]}$	$cm11$ 13.1.61	$\{E,\ M_{[210]}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p31m'$ 70.3.478	$\{E,\ C_{3z},\ C_{3z}^{-1},\ M_{[210]}\mathcal{T},\ M_{[120]}\mathcal{T},\ M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$cm'11$ 13.3.63	$\{E,\ M_{[120]}\mathcal{T}\}$	0	v
			$\mathrm{PL}^*_{(100)}$	$cm'11$ 13.3.63	$\{E,\ M_{[210]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{[1\bar{1}0]}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
71 71.1.479 $p^1\bar{3}^11^1m$ SI p. 31	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}^*_{[1\bar{1}0]}$	$c2/m11$ 18.1.95	$\{E, C_{2,[1\bar{1}0]}, \mathcal{P}, M_{[1\bar{1}0]}\}$	–	–
			$\mathrm{AX}_{[120]}$	$c2/m11$ 18.1.95	$\{E, C_{2,[120]}, \mathcal{P}, M_{[120]}\}$	–	0
			$\mathrm{AX}^*_{[210]}$	$c2/m11$ 18.1.95	$\{E, C_{2,[210]}, \mathcal{P}, M_{[210]}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p\bar{3}1m'$ 71.4.482	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[210]}\mathcal{T}, C_{2,[120]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}, \bar{C}_{3z}, \bar{C}_{3z}^{-1}, M_{[210]}\mathcal{T}, M_{[120]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[120]}\mathcal{T}, \mathcal{P}, M_{[120]}\mathcal{T}\}$	0	–
			$\mathrm{PL}^*_{(100)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[210]}\mathcal{T}, \mathcal{P}, M_{[210]}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}, M_{[1\bar{1}0]}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
72 72.1.484 $p^1\bar{3}^1m^11$ SI p. 31	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$	$\mathrm{AX}_{[100]}$	$c2/m11$ 18.1.95	$\{E, C_{2,[100]}, \mathcal{P}, M_{[100]}\}$	0	–
			$\mathrm{AX}^*_{[010]}$	$c2/m11$ 18.1.95	$\{E, C_{2,[010]}, \mathcal{P}, M_{[010]}\}$	–	–
			$\mathrm{AX}^*_{[110]}$	$c2/m11$ 18.1.95	$\{E, C_{2,[110]}, \mathcal{P}, M_{[110]}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p\bar{3}m'1$ 72.4.487	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[110]}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[010]}\mathcal{T}, \mathcal{P}, \bar{C}_{3z}, \bar{C}_{3z}^{-1}, M_{[110]}\mathcal{T}, M_{[100]}\mathcal{T}, M_{[010]}\mathcal{T}\}$	0	0
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[110]}\mathcal{T}, \mathcal{P}, M_{[110]}\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[100]}\mathcal{T}, \mathcal{P}, M_{[100]}\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[010]}\mathcal{T}, \mathcal{P}, M_{[010]}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
73 73.1.489 p^16 SI p. 31	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$	$\mathrm{AX}_{[001]}$	$p6$ 73.1.489	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}, C_{6z}^{-1}, C_{6z}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
74 74.1.492 $p^1\bar{6}$ SI p. 31	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$	$\mathrm{AX}_{[001]}$	$p\bar{6}$ 74.1.492	$\{E, C_{3z}, C_{3z}^{-1}, M_z, \bar{C}_{6z}^{-1}, \bar{C}_{6z}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11m'$ 4.3.14	$\{E, M_z\mathcal{T}\}$	–	–
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
75 75.1.495 $p^16/^1m$ SI p. 31	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$	$\mathrm{AX}_{[001]}$	$p6/m$ 75.1.495	$\{E,\, C_{3z},\, C_{3z}^{-1},\, C_{2z},\, C_{6z}^{-1},\, C_{6z},\, \mathcal{P},\, \bar{C}_{3z},\, \bar{C}_{3z}^{-1},\, M_z,\, \bar{C}_{6z}^{-1},\, \bar{C}_{6z}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E,\, C_{2z}\mathcal{T},\, \mathcal{P},\, M_z\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E,\, \mathcal{P}\}$	–	–
76 76.1.500 $p^16^12^12$ SI p. 32	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$c22'2'$ 22.4.125	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[100]},\, C_{2,[120]}\mathcal{T}\}$	+	–
			$\mathrm{AX}^*_{[010]}$	$c22'2'$ 22.4.125	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[010]},\, C_{2,[210]}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[110]},\, C_{2,[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[110]}\mathcal{T},\, C_{2,[1\bar{1}0]}\}$	v	v
			$\mathrm{AX}_{[120]}$	$c22'2'$ 22.4.125	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[100]}\mathcal{T},\, C_{2,[120]}\}$	–	+
			$\mathrm{AX}^*_{[210]}$	$c22'2'$ 22.4.125	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[010]}\mathcal{T},\, C_{2,[210]}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p62'2'$ 76.3.502	$\{E,\, C_{3z},\, C_{3z}^{-1},\, C_{2z},\, C_{6z}^{-1},\, C_{6z},\, C_{2,[110]}\mathcal{T},\, C_{2,[100]}\mathcal{T},\, C_{2,[010]}\mathcal{T},\, C_{2,[210]}\mathcal{T},\, C_{2,[120]}\mathcal{T},\, C_{2,[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11\bar{2}'$ 3.3.10	$\{E,\, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$c2'11$ 10.3.47	$\{E,\, C_{2,[120]}\mathcal{T}\}$	+	–
			$\mathrm{PL}^*_{(100)}$	$c2'11$ 10.3.47	$\{E,\, C_{2,[210]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$c2'11$ 10.3.47	$\{E,\, C_{2,[110]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E,\, C_{2,[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'11$ 10.3.47	$\{E,\, C_{2,[100]}\mathcal{T}\}$	–	+
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2'11$ 10.3.47	$\{E,\, C_{2,[010]}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
77 (S1) 77.1.505 $p^16^1m^1m$ SI p. 32	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cmm'2'$ 26.4.150	$\{E,\, C_{2z}\mathcal{T},\, M_{[100]},\, M_{[120]}\mathcal{T}\}$	0	v
			$\mathrm{AX}^*_{[010]}$	$cmm'2'$ 26.4.150	$\{E,\, C_{2z}\mathcal{T},\, M_{[010]},\, M_{[210]}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E,\, C_{2z}\mathcal{T},\, M_{[110]},\, M_{[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E,\, C_{2z}\mathcal{T},\, M_{[110]}\mathcal{T},\, M_{[1\bar{1}0]}\}$	v	v
			$\mathrm{AX}_{[120]}$	$cmm'2'$ 26.4.150	$\{E,\, C_{2z}\mathcal{T},\, M_{[100]}\mathcal{T},\, M_{[120]}\}$	v	0
			$\mathrm{AX}^*_{[210]}$	$cmm'2'$ 26.4.150	$\{E,\, C_{2z}\mathcal{T},\, M_{[010]}\mathcal{T},\, M_{[210]}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p6m'm'$ 77.3.507	$\{E,\, C_{3z},\, C_{3z}^{-1},\, C_{2z},\, C_{6z}^{-1},\, C_{6z},\, M_{[110]}\mathcal{T},\, M_{[100]}\mathcal{T},\, M_{[010]}\mathcal{T},\, M_{[210]}\mathcal{T},\, M_{[120]}\mathcal{T},\, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11'$ 3.3.10	$\{E,\, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$cm'11$ 13.3.63	$\{E,\, M_{[120]}\mathcal{T}\}$	0	v
			$\mathrm{PL}^*_{(100)}$	$cm'11$ 13.3.63	$\{E,\, M_{[210]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E,\, M_{[110]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\, M_{[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\, M_{[100]}\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$cm'11$ 13.3.63	$\{E,\, M_{[010]}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
78 78.1.510 $p^1\bar{6}^1m^12$ SI p. 32	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ M_{[100]},\ C_{2,[120]}\mathcal{T}\}$	0	–
		$\mathrm{AX}^*_{[010]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ M_{[010]},\ C_{2,[210]}\mathcal{T}\}$	–	–
		$\mathrm{AX}^*_{[110]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ M_{[110]},\ C_{2,[1\bar{1}0]}\mathcal{T}\}$	–	–
		$\mathrm{AX}^*_{[1\bar{1}0]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ M_{[110]}\mathcal{T},\ C_{2,[1\bar{1}0]}\}$	–	–
		$\mathrm{AX}_{[120]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ M_{[100]}\mathcal{T},\ C_{2,[120]}\}$	–	0
		$\mathrm{AX}^*_{[210]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ M_{[010]}\mathcal{T},\ C_{2,[210]}\}$	–	–
		$\mathrm{AX}_{[001]}$	$p\bar{6}m'2'$ 78.3.512	$\{E,\ C_{3z},\ C_{3z}^{-1},\ M_z,\ \bar{C}_{6z}^{-1},\ \bar{C}_{6z},\ M_{[110]}\mathcal{T},\ M_{[100]}\mathcal{T},\ M_{[010]}\mathcal{T},\ C_{2,[210]}\mathcal{T},\ C_{2,[120]}\mathcal{T},\ C_{2,[1\bar{1}0]}\mathcal{T}\}$	0	0
		$\mathrm{PL}_{(001)}$	$p11m'$ 4.3.14	$\{E,\ M_z\mathcal{T}\}$	–	–
		$\mathrm{PL}_{(010)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[120]}\mathcal{T}\}$	+	–
		$\mathrm{PL}^*_{(100)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[210]}\mathcal{T}\}$	v	v
		$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E,\ M_{[110]}\mathcal{T}\}$	v	v
		$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[1\bar{1}0]}\mathcal{T}\}$	v	v
		$\mathrm{PL}_{(2\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{[100]}\mathcal{T}\}$	v	0
		$\mathrm{PL}^*_{(1\bar{2}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{[010]}\mathcal{T}\}$	v	v
		GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
79 79.1.515 $p^1\bar{6}^12^1m$ SI p. 33	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ C_{2,[100]},\ M_{[120]}\mathcal{T}\}$	0	–
		$\mathrm{AX}^*_{[010]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ C_{2,[010]},\ M_{[210]}\mathcal{T}\}$	–	–
		$\mathrm{AX}^*_{[110]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ C_{2,[110]},\ M_{[1\bar{1}0]}\mathcal{T}\}$	–	–
		$\mathrm{AX}^*_{[1\bar{1}0]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ C_{2,[110]}\mathcal{T},\ M_{[1\bar{1}0]}\}$	–	–
		$\mathrm{AX}_{[120]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ C_{2,[100]}\mathcal{T},\ M_{[120]}\}$	–	0
		$\mathrm{AX}^*_{[210]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ C_{2,[010]}\mathcal{T},\ M_{[210]}\}$	–	–
		$\mathrm{AX}_{[001]}$	$p\bar{6}2'm'$ 79.3.517	$\{E,\ C_{3z},\ C_{3z}^{-1},\ M_z,\ \bar{C}_{6z}^{-1},\ \bar{C}_{6z},\ C_{2,[110]}\mathcal{T},\ C_{2,[100]}\mathcal{T},\ C_{2,[010]}\mathcal{T},\ M_{[210]}\mathcal{T},\ M_{[120]}\mathcal{T},\ M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
		$\mathrm{PL}_{(001)}$	$p11m'$ 4.3.14	$\{E,\ M_z\mathcal{T}\}$	–	–
		$\mathrm{PL}_{(010)}$	$cm'11$ 13.3.63	$\{E,\ M_{[120]}\mathcal{T}\}$	0	v
		$\mathrm{PL}^*_{(100)}$	$cm'11$ 13.3.63	$\{E,\ M_{[210]}\mathcal{T}\}$	v	v
		$\mathrm{PL}^*_{(110)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[110]}\mathcal{T}\}$	v	v
		$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{[1\bar{1}0]}\mathcal{T}\}$	v	v
		$\mathrm{PL}_{(2\bar{1}0)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[100]}\mathcal{T}\}$	–	+
		$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[010]}\mathcal{T}\}$	v	v
		GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Magnetic-axis direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
80 80.1.520 $p^16/^1m^1m^1m$ SI p. 33	FM/FIM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cmm'm'$ 47.5.334	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[100]},\, C_{2,[120]}\mathcal{T},\, \mathcal{P},\, M_z\mathcal{T},\, M_{[100]},\, M_{[120]}\mathcal{T}\}$	0	–
			$\mathrm{AX}^*_{[010]}$	$cmm'm'$ 47.5.334	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[010]},\, C_{2,[210]}\mathcal{T},\, \mathcal{P},\, M_z\mathcal{T},\, M_{[010]},\, M_{[210]}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[110]}$	$cmm'm'$ 47.5.334	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[110]},\, C_{2,[1\bar{1}0]}\mathcal{T},\, \mathcal{P},\, M_z\mathcal{T},\, M_{[110]},\, M_{[1\bar{1}0]}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[110]}\mathcal{T},\, C_{2,[1\bar{1}0]},\, \mathcal{P},\, M_z\mathcal{T},\, M_{[110]}\mathcal{T},\, M_{[1\bar{1}0]}\}$	–	–
			$\mathrm{AX}_{[120]}$	$cmm'm'$ 47.5.334	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[100]}\mathcal{T},\, C_{2,[120]},\, \mathcal{P},\, M_z\mathcal{T},\, M_{[100]}\mathcal{T},\, M_{[120]}\}$	–	0
			$\mathrm{AX}^*_{[210]}$	$cmm'm'$ 47.5.334	$\{E,\, C_{2z}\mathcal{T},\, C_{2,[010]}\mathcal{T},\, C_{2,[210]},\, \mathcal{P},\, M_z\mathcal{T},\, M_{[010]}\mathcal{T},\, M_{[210]}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p6/mm'm'$ 80.4.523	$\{E,\, C_{3z},\, C_{3z}^{-1},\, C_{2z},\, C_{6z}^{-1},\, C_{6z},\, C_{2,[110]}\mathcal{T},\, C_{2,[100]}\mathcal{T},\, C_{2,[010]}\mathcal{T},\, C_{2,[210]}\mathcal{T},\, C_{2,[120]}\mathcal{T},\, C_{2,[1\bar{1}0]}\mathcal{T},\, \mathcal{P},\, \bar{C}_{3z},\, \bar{C}_{3z}^{-1},\, M_z,\, \bar{C}_{6z}^{-1},\, \bar{C}_{6z},\, M_{[110]}\mathcal{T},\, M_{[100]}\mathcal{T},\, M_{[010]}\mathcal{T},\, M_{[210]}\mathcal{T},\, M_{[120]}\mathcal{T},\, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E,\, C_{2z}\mathcal{T},\, \mathcal{P},\, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$c2'/m'11$ 18.4.98	$\{E,\, C_{2,[120]}\mathcal{T},\, \mathcal{P},\, M_{[120]}\mathcal{T}\}$	0	–
			$\mathrm{PL}^*_{(100)}$	$c2'/m'11$ 18.4.98	$\{E,\, C_{2,[210]}\mathcal{T},\, \mathcal{P},\, M_{[210]}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E,\, C_{2,[110]}\mathcal{T},\, \mathcal{P},\, M_{[110]}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E,\, C_{2,[1\bar{1}0]}\mathcal{T},\, \mathcal{P},\, M_{[1\bar{1}0]}\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E,\, C_{2,[100]}\mathcal{T},\, \mathcal{P},\, M_{[100]}\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2'/m'11$ 18.4.98	$\{E,\, C_{2,[010]}\mathcal{T},\, \mathcal{P},\, M_{[010]}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E,\, \mathcal{P}\}$	–	–

Appendix A. Complete coupled-component constraints

$A^\ell = \chi^\ell_{xyy}$, $B^\ell = \chi^\ell_{yxx}$. Each row below gives the entire allowed linear space for a starred branch, with independent real parameters u or u_1, u_2. These relations supplement the $+, -, 0, v$ flags; they are not additional assumptions.

No.	cSLG label	Magnetic-axis direction	Complete allowed space
53	53.1.374	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ 1,\ 1,\ 0) + u_2(1,\ 0,\ 0,\ 1)$
53	53.1.374	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ -1,\ 1,\ 0) + u_2(-1,\ 0,\ 0,\ 1)$
53	53.1.374	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ -1,\ 1,\ 0) + u_2(-1,\ 0,\ 0,\ 1)$
53	53.1.374	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ 1,\ 1,\ 0) + u_2(1,\ 0,\ 0,\ 1)$
54	54.1.381	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ 1,\ 1,\ 0) + u_2(1,\ 0,\ 0,\ 1)$
54	54.1.381	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ -1,\ 1,\ 0) + u_2(-1,\ 0,\ 0,\ 1)$
54	54.1.381	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ -1,\ 1,\ 0) + u_2(-1,\ 0,\ 0,\ 1)$
54	54.1.381	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ 1,\ 1,\ 0) + u_2(1,\ 0,\ 0,\ 1)$
55	55.1.386	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(-1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ -1,\ 1)$
55	55.1.386	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ 1,\ 1)$
55	55.1.386	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ 1,\ 1)$
55	55.1.386	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(-1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ -1,\ 1)$
56	56.1.393	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(-1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ -1,\ 1)$
56	56.1.393	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ 1,\ 1)$
56	56.1.393	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ 1,\ 1)$
56	56.1.393	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(-1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ -1,\ 1)$
57	57.1.398	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(-1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ -1,\ 1)$
57	57.1.398	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ 1,\ 1)$
57	57.1.398	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ 1,\ 1)$
57	57.1.398	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(-1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ -1,\ 1)$
58	58.1.405	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(-1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ -1,\ 1)$
58	58.1.405	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ 1,\ 1)$
58	58.1.405	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ 1,\ 1)$
58	58.1.405	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(-1,\ 1,\ 0,\ 0) + u_2(0,\ 0,\ -1,\ 1)$
59	59.1.410	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ 1,\ 1,\ 0) + u_2(1,\ 0,\ 0,\ 1)$
59	59.1.410	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ -1,\ 1,\ 0) + u_2(-1,\ 0,\ 0,\ 1)$
59	59.1.410	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ -1,\ 1,\ 0) + u_2(-1,\ 0,\ 0,\ 1)$
59	59.1.410	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ 1,\ 1,\ 0) + u_2(1,\ 0,\ 0,\ 1)$
60	60.1.417	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ 1,\ 1,\ 0) + u_2(1,\ 0,\ 0,\ 1)$
60	60.1.417	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ -1,\ 1,\ 0) + u_2(-1,\ 0,\ 0,\ 1)$
60	60.1.417	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ -1,\ 1,\ 0) + u_2(-1,\ 0,\ 0,\ 1)$
60	60.1.417	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0,\ 1,\ 1,\ 0) + u_2(1,\ 0,\ 0,\ 1)$
61	61.1.422	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u(1,\ -1,\ -1,\ 1)$
61	61.1.422	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u(-1,\ -1,\ 1,\ 1)$
61	61.1.422	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u(-1,\ -1,\ 1,\ 1)$
61	61.1.422	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u(1,\ -1,\ -1,\ 1)$
62	62.1.435	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u(1,\ -1,\ -1,\ 1)$
62	62.1.435	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u(-1,\ -1,\ 1,\ 1)$
62	62.1.435	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u(-1,\ -1,\ 1,\ 1)$
62	62.1.435	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u(1,\ -1,\ -1,\ 1)$
63	63.1.444	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u(1,\ -1,\ -1,\ 1)$
63	63.1.444	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u(-1,\ -1,\ 1,\ 1)$
63	63.1.444	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u(-1,\ -1,\ 1,\ 1)$
63	63.1.444	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u(1,\ -1,\ -1,\ 1)$
64	64.1.453	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u(1,\ -1,\ -1,\ 1)$
64	64.1.453	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u(-1,\ -1,\ 1,\ 1)$

No.	cSLG label	Magnetic-axis direction	Complete allowed space
64	64.1.453	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ -1,\ 1,\ 1\big)$
64	64.1.453	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ -1,\ -1,\ 1\big)$
67	67.1.467	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
67	67.1.467	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
67	67.1.467	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
67	67.1.467	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
68	68.1.470	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
68	68.1.470	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
68	68.1.470	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
68	68.1.470	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
69	69.1.473	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
69	69.1.473	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
69	69.1.473	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
69	69.1.473	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
70	70.1.476	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
70	70.1.476	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
70	70.1.476	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
70	70.1.476	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
71	71.1.479	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -3,\ \sqrt{3},\ 3\big)$
71	71.1.479	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -3,\ -\sqrt{3},\ 3\big)$
71	71.1.479	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -1,\ \sqrt{3},\ 1\big)$
71	71.1.479	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -1,\ -\sqrt{3},\ 1\big)$
72	72.1.484	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -1,\ \sqrt{3},\ 1\big)$
72	72.1.484	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -1,\ -\sqrt{3},\ 1\big)$
72	72.1.484	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -3,\ \sqrt{3},\ 3\big)$
72	72.1.484	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -3,\ -\sqrt{3},\ 3\big)$
76	76.1.500	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
76	76.1.500	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
76	76.1.500	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
76	76.1.500	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
76	76.1.500	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
76	76.1.500	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
76	76.1.500	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
76	76.1.500	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
77	77.1.505	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
77	77.1.505	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
77	77.1.505	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
77	77.1.505	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
77	77.1.505	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
77	77.1.505	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
77	77.1.505	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
77	77.1.505	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
78	78.1.510	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -1,\ \sqrt{3},\ 1\big)$
78	78.1.510	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -1,\ -\sqrt{3},\ 1\big)$
78	78.1.510	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -3,\ \sqrt{3},\ 3\big)$
78	78.1.510	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -3,\ -\sqrt{3},\ 3\big)$
78	78.1.510	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
78	78.1.510	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
78	78.1.510	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$

No.	cSLG label	Magnetic-axis direction	Complete allowed space
78	78.1.510	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\, 3,\, 0,\, 0\big) + u_2\big(0,\, 0,\, -\sqrt{3},\, 3\big)$
79	79.1.515	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -1,\, \sqrt{3},\, 1\big)$
79	79.1.515	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -1,\, -\sqrt{3},\, 1\big)$
79	79.1.515	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -3,\, \sqrt{3},\, 3\big)$
79	79.1.515	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -3,\, -\sqrt{3},\, 3\big)$
79	79.1.515	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\, 1,\, 0,\, 0\big) + u_2\big(0,\, 0,\, \sqrt{3},\, 1\big)$
79	79.1.515	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\, -\sqrt{3},\, 2,\, 0\big) + u_2\big(-\sqrt{3},\, -1,\, 0,\, 2\big)$
79	79.1.515	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\, 1,\, 0,\, 0\big) + u_2\big(0,\, 0,\, -\sqrt{3},\, 1\big)$
79	79.1.515	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\, \sqrt{3},\, 2,\, 0\big) + u_2\big(\sqrt{3},\, -1,\, 0,\, 2\big)$
80	80.1.520	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -1,\, \sqrt{3},\, 1\big)$
80	80.1.520	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -1,\, -\sqrt{3},\, 1\big)$
80	80.1.520	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -3,\, \sqrt{3},\, 3\big)$
80	80.1.520	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -3,\, -\sqrt{3},\, 3\big)$
80	80.1.520	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -1,\, \sqrt{3},\, 1\big)$
80	80.1.520	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -3,\, \sqrt{3},\, 3\big)$
80	80.1.520	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -1,\, -\sqrt{3},\, 1\big)$
80	80.1.520	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -3,\, -\sqrt{3},\, 3\big)$